

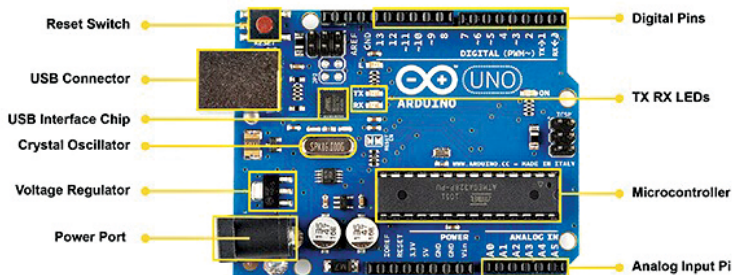
An Arduino consists of a microcontroller or physical programmable circuit board, along with software to load your codes on to the microcontroller. It runs on C++. Arduino has a standard form factor irrespective of where it is operating. The codes are pretty simple and easy to load. Arduino has several models dedicated to Entry Level (Basic processing), Enhanced Performers, IoT, Wearable, and Education.

In this section, we will discuss the components that we have on an Arduino board. Let's begin.

Components of Arduino

The Microcontroller

The microcontroller is the head of the Arduino, and it does what you ask it to do. There are various microcontrollers used in different Arduino boards. Let us discuss the ATmega328 present in UNO R3. ATmega328 belongs to the AVR family and is an 8-bit device. It means that its internal registers



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and data bus can handle 8 signals simultaneously without overlapping. It also has three types of memories to keep it up and running Flash Memory, SRAM memory, and EEPROM memory.

The Flash memory is a 32KB non-volatile memory used for storing application. It is the primary reason why you get the same software whenever you plug in the board.

The SRAM memory is a 2KB volatile memory and is used to store the application variables when it is operating.

The EEPROM memory is a minute 1KB non-volatile memory to store the data that will be useful even after the board is switched off. The data stored on it will be available whenever you boot your board.